Selective Recovery for Multilingual ASR: Candidate Acquisition and Acceptance

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***Abstract*—We separate fallback acquisition from acceptance using a compression trigger and a fixed four-feature selector. On a locked confirmation panel of 7,500 Common Voice 27.0 (CV27) recordings, selective recovery lowers macro word error rate (WER) by 2.26 percentage points (pp) over baseline rescoring and matches unconditional recovery's error counts in every seed. Retrospective CV27 TEST comparisons show 99.98% retention of macro-weighted positive edit gain. Reselection improves unscreened fallbacks; direct adoption can outperform the fixed selector on internally screened Hugging Face (HF) outputs. Matching acceptance to HF banks with edit-gap weighting yields the lowest exploratory CV27 WER of 31.44%. Across three checkpoints, selective acquisition saves up to 5.15% latency within reranked pipelines. Native HF remains faster but incurs 1.46 pp higher WER on FLEURS and over 2.1 pp on CV27. Code is available at https://github.com/Zhifan-Pan/ISCAS2027.**

***Index Terms*—multilingual speech recognition, selective recovery, candidate acquisition, N-best reranking, confidence estimation, conditional computation**

I. Introduction

Multilingual ASR encounters routine errors and catastrophic repetition loops. Additional decoding helps only when it produces a better candidate that the scoring rule accepts. Separating acquisition from acceptance reveals when recovery helps or wastes computation.

Multilingual N-best reranking [1], [2], discriminative rescoring [3]–[5], voting [6] and confusion networks [7] exploit existing hypotheses. Generative correction expands banks [8]–[11], and risk-based pruning and ASR sampling allocate computation [12], [13]. Conditional computation uses recognizer cascades [14], selective passes [15], encoder proxies [16], selective correction [17] or language-specific distillation [18], but confidence need not predict the benefit of deferral [19].

We distinguish routine beam rescoring from sparse repetition recovery. We ask when bank expansion helps and when reselection helps or harms a screened fallback. The locked panel isolates acquisition; retrospective controls examine acceptance. We compare expected risk, pairwise ranking and minimum Bayes risk on matched banks [20]–[22]. These controls isolate acquisition and acceptance gains within reranked pipelines. Native HF trades higher WER for faster decoding.

II. Controlled Acquisition and Acceptance

A. Recording-level acquisition

We normalize and deduplicate five beams per window i into B_i, retaining first occurrences and sources. With raw top beam b_i and fallback g_i, $F_i = B_i \cup \{g_i\}$ contains at most six candidates. Before recovery, $\mathrm{CR}_{\mathrm{text}}$ divides top-beam UTF-8 byte length by zlib-compressed length, with zero for empty strings. The gate $z_i = \mathbf{1}\{\mathrm{CR}_{\mathrm{text},i} > 2.4\}$ uses Whisper's standard text threshold [29], fixed before confirmation. Recording-level invocation $z^{\mathrm{rec}} = \max_i z_i$ expands every window, even singletons (Fig. 1), as a conservative control. On single-window CV27 audio, $z^{\mathrm{rec}} = z_i$; Section IV-D examines local routing for multi-window FLEURS.

Recovery follows Hugging Face temperatures $0, 0.2, \ldots, 1$ with compression, log-probability and no-speech thresholds 1.35, -1 and 0.6. The internal check compresses serialized token-ID bytes [23] with a stricter retry threshold; the outer gate screens decoded text for severe loops. Shared leading bytes in two-byte Arabic/Persian characters can increase compression relative to ASCII Latin text at similar character repetition. Native HF uses the same retry configuration without N-best rescoring.

B. Four-feature shared acceptance

For features $f(y) = (a, p, t, r)$, a is the mean teacher-forced log probability over T body tokens without prefix, padding or end tokens, and $t = \log(1 + T)$. XLM-R-base pseudo-log-likelihood p [24], [25], or PLL, averages individually masked nonspecial subword log probabilities in centered contexts of at most 510 subwords of the original string, rather than within-word masking [26]. Empty outputs have no body tokens or PLL targets and map to $f(\emptyset) = (0, 0, 0, 0)$, but normalization alone need not yield zero features.

For normalized words $w_{1:M}$, the sequence-wide overlapping count is $c_y(u) = \sum_{k=1}^{M-n+1} \mathbf{1}\{w_{k:k+n-1} = u\}$. We measure repetition by

$$r_{n,h}(y) = \frac{\sum_{j=1}^{M-n+1} \mathbf{1}\{c_y(w_{j:j+n-1}) \geq h\}}{\max(1, M - n + 1)}. \tag{1}$$

We use $r(y) = r_{4,3}(y)$, which is zero for $M < 4$. Repetitive sequences also motivate unlikelihood training [27].

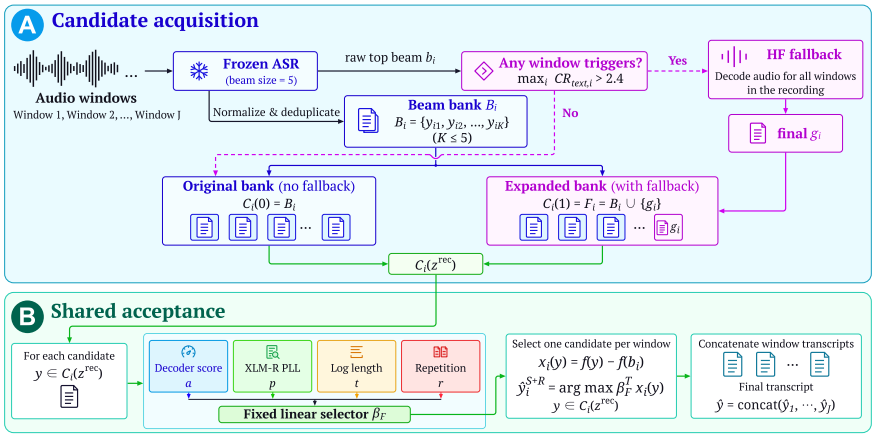


Fig. 1. Acquisition (A) expands every window's candidate bank if any window triggers. Acceptance (B) applies the same fixed four-feature selector to the original or expanded bank. Selected window transcripts are then concatenated.

Centering $x_i(y) = f(y) - f(b_i)$ preserves rankings and softmax probabilities. An HF-proposed empty scores $-\beta_F^\top f(b_i)$, an implicit abstention baseline. For $x_{ik} = x_i(y_{ik})$, root-mean-square (RMS) scales satisfy $s_d^2 = \sum_{\mathcal{D}} \omega_{ik} x_{ik,d}^2 / \sum_{\mathcal{D}} \omega_{ik}$ over non-top training-fold pairs $\mathcal{D} = \{(i,k) : i \in \bigcup_{\ell=1}^{L} \mathcal{I}_\ell,\ y_{ik} \in C_i \setminus \{b_i\}\}$. G4 uses $\omega_{ik} = 1/[N_\ell(|C_i| - 1)]$; all HF refits use $\omega_{ik} = 1$. Scales below 10^{-12} become one. For reference y_i^* and word-edit count E, define $\Delta_{ik} = E(b_i, y_i^*) - E(y_{ik}, y_i^*)$ and $\pi_{ik} = \text{softmax}_k(\theta^\top \text{diag}(s)^{-1} x_{ik})$. Both softmax and loss include all candidates, with $x_i(b_i) = \mathbf{0}$ and $\Delta_{i,b_i} = 0$. Expected-risk fitting [20] minimizes

$$\mathcal{L}(\theta) = -\frac{1}{L} \sum_{\ell=1}^{L} \frac{\sum_{i \in \mathcal{I}_\ell} \sum_{k=1}^{|C_i|} \pi_{ik} \Delta_{ik}}{N_\ell} + \lambda \|\theta\|_2^2. \quad (2)$$

Here $\mathcal{I}_\ell = \{i : \text{lang}(i) = \ell\}$ indexes fitting windows, N_ℓ counts reference words and $L = 5$.

The shared weights $\beta_F = \text{diag}(s)^{-1}\theta$ are fitted per run on greedy-expanded DEV and fixed across sources, with ties broken by bank order. With $\hat{y}_i^B = \text{argmax}_{y \in B_i} \beta_F^\top x_i(y)$, $C_i(0) = B_i$ and $C_i(1) = F_i$, the policies are

$$\hat{y}_i^{S+R} = \underset{y \in C_i(z^{\text{rec}})}{\text{argmax}}\ \beta_F^\top x_i(y),$$
$$\hat{y}_i^{\text{Direct}} = \begin{cases} \hat{y}_i^B, & z^{\text{rec}} = 0, \\ g_i, & z^{\text{rec}} = 1. \end{cases} \quad (3)$$

Policy B reranks B_i, and N+R requests and reranks F_i unconditionally. S+R and Direct share acquisition and use B off-trigger, but Direct bypasses selection and PLL on triggers. Window transcripts are concatenated for full-reference scoring.

C. Matched acceptance controls

Table I lists policies; Table II compares acceptors. G4 stays frozen. We fit H4 by expected risk on HF-expanded DEV banks and H3 without PLL, using (a, t, r). RankNet [21] minimizes

$$\mathcal{L}_{\text{rank}} = \frac{1}{L} \sum_{\ell=1}^{L} \frac{\sum_{i \in \mathcal{I}_\ell} \ell_i}{N_\ell} + \lambda \|\theta\|_2^2,$$

with $\ell_i = |\mathcal{P}_i|^{-1} \sum_{(j,k) \in \mathcal{P}_i} w_{jk} \log(1 + e^{-\beta^\top (x_{ij} - x_{ik})})$ over unequal-edit pairs $\mathcal{P}_i = \{(j,k) : E_j < E_k\}$, where $E_j = E(y_{ij}, y_i^*)$ and $\beta = \text{diag}(s)^{-1}\theta$. Empty $\mathcal{P}_i$ give $\ell_i = 0$. HR4 uses $w_{jk} = 1$; HWR4 uses $|E_j - E_k|/c$, with c the training-fold mean gap under HR4's pair weights.

TABLE I
WER (%) AND PP CONTRASTS BEFORE ROUNDING. CHECKS DENOTE HF ACQUISITION, CR GATING AND R RESELECTION (†: OFF-TRIGGER ONLY).

Path	HF	CR	R	Scorer	FLEURS	CV27
Raw beam-5 top-1	–	–	–	–	18.90	36.41
Native HF [23]	✓	–	–	–	18.55	33.80
B	–	–	✓	G4	17.45	33.98
N+R	✓	–	✓	G4	17.09	31.68
S+R	✓	✓	✓	G4	17.09	31.68
Direct	✓	✓	✓†	G4	17.09	31.46
H3Direct	✓	✓	✓†	H3	18.68	33.64
A. Reference-assisted TEST oracle bounds						
Original-bank oracle					16.19	30.19
Full-HF oracle					15.83	27.68
CR-routed oracle					15.83	27.68
S+R gap to routed oracle, pp					1.26	4.00

B. Locked CV27 confirmation (7,500 recordings)			
	B	N+R = S+R	Gain
Macro-average	35.02	32.75	2.26
Contrast		ΔWER	95% upper bound
S+R − B		-2.26	-1.32
S+R − N+R		0.00	0.00

MBR [22] selects $\hat{y}_i = \text{argmin}_{y \in C_i} \sum_{y' \in C_i} q_i(y') E(y, y')$. U-MBR uses $q_i(y') \propto 1$ and D-MBR uses $q_i(y') = \text{softmax}(Ta/\tau)_{y'}$. DEV WER selects $\tau \in \{0.1, 0.3, 1, 3, 10\}$, favoring larger values on ties; candidate ties retain bank order. *Complete* policies use each acceptor on both banks; *triggered* comparisons keep G4 off-trigger with full-panel denominators.

III. Data and Evaluation

We adapt Whisper-small [29] on 44.60 h of FLEURS TRAIN [28] across Arabic, Persian, Polish, Turkish and Italian using LoRA [30] over three random seeds (4,656 updates, rank and scale four, zero dropout across all attention projections).

LoRA training excludes CV27 TRAIN. Selectors are tuned on CV27 DEV, where recovery was checked with the fixed gate before confirmation. Evaluation on FLEURS TEST (13.07 h; 3,665 recordings) and CV27 TEST (83.00 h; 58,514 recordings) [31] is retrospective because both splits were accessed during development.

The locked confirmation panel has 7,500 CV27 TRAIN recordings (1,500 per language) with no previously accessed audio or reference texts. Predictions, stopping rules and the one-sided 0.10 pp noninferiority margin were locked before reference evaluation. We use 10,000 synchronized bootstrap draws of speaker components within languages. Speakers recurring across splits limit generalization to this sampling frame. Confirmation covers the original G4 acquisition policy; later gates, refits and window policies remain retrospective.

We weight costs by inverse sampling probabilities for 255 FLEURS and 472 CV27 recordings. The sample covers all first-checkpoint triggers or Native HF output changes and ten ordinary recordings per language and duration quintile. Study A measures four paths across three checkpoints; Study B measures seven on one. Later-checkpoint omissions comprise five FLEURS and 207 CV27 triggers (0.05% and 0.12% of seed–recordings). Triggered N+R and S+R execute identical operations, though sampling shares do not bound timing variation. Study B covers all its triggers and corroborates Study A's savings (5.18% and 4.54% versus 5.15% and 4.57%). Both use three randomized paired rounds with each recording block on one of four NVIDIA A40 (48 GB) GPUs, resident models, within-path encoder and first-search reuse, and 32-mask PLL batches. Waveform-to-output timing excludes loading, warmup and interfered blocks. Peak CUDA memory is the maximum per-process allocation in MiB over calls and rounds. Session-conditional intervals use 4,000 paired stratified draws with finite-population correction; Study A averages checkpoints with synchronized draws.

Known-language decoding uses bfloat16, five beams, length penalty one, native suppression and a 444-token cap in Transformers 4.57.6. At 16 kHz, N samples give $J = \lceil N/480000 \rceil$ windows bounded by $\lfloor jN/J \rfloor$ without previous-window text. Deduplication, WER and r share NFC normalization, digit unification, punctuation-to-space, Persian mapping and Turkish casing [32], with fixed source RNG mappings. WER averages five language aggregates and three seeds; calls pool seed–recordings. TEST intervals use 2,000 paired client/reference-component draws within languages, synchronized across seeds with fits and source draws frozen. FLEURS comparisons are descriptive because component metadata are unverified.

IV. Results

A. Acquisition with fixed acceptance

On the locked confirmation panel, S+R gains 2.26 pp over B, with seed gains of 0.20 to 4.54 pp (Table I). Matching N+R's error counts in every seed meets the prespecified 0.10 pp margin but does not guarantee zero loss under distribution shift. Of 61 triggers, 49 improve edits and two worsen them.

TABLE II
Complete-policy TEST WER (%) and gain B−S (pp) on HF banks. Checks denote PLL, pairwise P and edit-gap weighting W.

Acceptor	PLL	P	W	FLEURS			CV27		
				B	S	Gain	B	S	Gain
G4 [20]	✓	–	–	17.45	17.09	0.37	33.98	31.68	2.30
H4 [20]	✓	–	–	17.46	17.09	0.37	33.97	31.52	2.45
HR4 [21]	✓	✓	–	17.51	17.14	0.37	34.02	32.30	1.72
H3 [20]	–	–	–	19.05	18.68	0.37	36.15	33.77	2.38
HWR4 [21]	✓	✓	✓	17.47	17.10	0.37	33.97	31.44	2.53
U-MBR [22]	–	–	–	19.14	19.14	0.00	36.91	36.88	0.03
D-MBR [22]	–	–	–	18.92	18.75	0.17	36.09	34.48	1.60
A. Original family: CV27 ΔWER (individual 97.5% CI)									
Triggered HR4 − H4				+0.72			[+0.40, +1.16]		
Complete H3 − H4				+2.25			[+2.16, +2.34]		
B. Later two-comparison family									
Triggered U-MBR − H4				+2.51			[+1.93, +3.18]		
Triggered HWR4 − H4				-0.08			[-0.15, -0.03]		

Retrospective evaluation yields 32.71% WER for Direct versus S+R's locked 32.75%. Both beat B and Native HF's retrospective 34.87% by over 2.1 pp. Direct's unrounded difference from S+R is -0.05 pp with a paired speaker-bootstrap 95% interval from -0.19 to $+0.04$ pp.

On TEST, S+R retains all ten beneficial FLEURS instances at 0.09% calls and 370 of 373 CV27 instances at 0.30%, missing five edits. After language-reference normalization and averaging languages and seeds, it retains 99.98% of positive edit gain. Normalized outputs differ on one FLEURS and 13 CV27 nontriggers despite nearly identical WER. N+R requests and selects an HF candidate there; S+R keeps B. Reference-assisted oracles minimize recording edits over concatenated original, expanded or CR-routed banks. Routing changes this bound by under 0.01 pp against S+R's 4.00 pp CV27 gap, leaving more headroom for acceptance.

B. Interaction with the fallback source

S+R rejects unscreened fallbacks and beats Direct by 0.53 pp with greedy recovery and 0.43 pp at temperature 0.2. HF already screens retries by compression, log-probability and no-speech criteria. Direct then leads G4 by 0.22 pp, with a 95% interval from 0.11 to 0.37 pp, concentrated in one checkpoint (-0.68 pp versus $+0.00$ and $+0.02$ pp). There, G4 retains repetitive beams whose decoder/PLL advantage outweighs their length and repetition penalties, rejecting better HF candidates. DEV favored HF reselection at 30.08% versus 30.09% and guided the fixed acquisition control.

Direct retains 67.13% of positive gain from unconditional final HF, compared with S+R's 99.98% from N+R. Relative to its own control, Direct forgoes 1.24 pp of benefit but avoids 3.58 pp of harm.

C. Acceptance on matched HF banks

On HF banks, H4 improves over G4 but trails Direct (31.52% versus 31.46% CV27 WER). HWR4 yields the lowest complete CV27 WER of 31.44% (Table II). Triggered HR4 trails H4 in every seed ($+0.91$, $+0.08$ and $+1.17$ pp). Edit-gap weighting reverses the mean contrast to -0.08 pp. Its 0.25 pp

TABLE III
POPULATION-WEIGHTED MS AND OBSERVED PEAK ALLOCATED MIB WITH BOTH MODELS RESIDENT. EACH STUDY USES THREE PAIRED ROUNDS.

Path	FLEURS		CV27	
	ms	MiB	ms	MiB
A. Four-path comparison across three checkpoints				
Native HF	1291.7	2980.8	517.4	2813.8
B	1352.7	3144.9	591.9	3133.1
N+R	1436.8	2988.3	639.2	3133.1
S+R	1362.8	3189.7	610.0	3133.1
B. Seven-path comparison on one checkpoint				
Native HF	1304.8	2978.4	531.2	2811.8
Raw top-1	1222.6	2946.6	479.9	2571.0
B	1364.0	3142.4	605.1	3131.1
N+R	1449.6	2986.3	657.9	3131.1
S+R	1374.5	3187.2	628.0	3131.1
Direct	1372.8	3187.2	625.8	2811.8
H3Direct	1314.0	3187.2	586.4	2811.8
Relative latency saving: S+R versus N+R (95% CI)				
Study A	5.15% [2.84, 9.41]		4.57% [4.40, 4.73]	
Study B	5.18% [2.88, 9.39]		4.54% [4.44, 4.66]	
Study B: differences between policies				
B − Raw top-1, ms	141.4		125.2	
N+R − B, ms	85.6		52.8	
S+R − B, ms	10.5		22.9	
Fallback increment saved	87.7%		56.6%	

gain in checkpoint two replaces 11 repetitive beams with lower-edit HF candidates. Intervals provide 95% Bonferroni coverage per family, with 10,000 synchronized draws for the later family. Repeating DEV selection with G4's RMS rule changes complete CV27 WER by $+0.044$, -0.057 and $+0.003$ pp for H4, HR4 and HWR4, respectively. Their ordering is unchanged.

FLEURS gains round to 0.37 pp for fitted acceptors because S equals its own B off-trigger. H3's B has two fewer errors on one of ten triggers, but S errors match and the average difference is below 0.006 pp. Nontriggers account for 97.23% of the refitted H3–H4 WER gap in routine selection, without isolating PLL's causal contribution or time share.

D. Cost, heterogeneity and supporting diagnostics

Within reranked pipelines, S+R preserves N+R's gain at lower cost (Table III). Study B's rescoring increments above raw top-1 are 141.4 ms and 125.2 ms on FLEURS and CV27. Selective acquisition saves 87.7% and 56.6% of fallback overhead. The unweighted 25-recording CV27 profile averages 85.9 ms teacher scoring and 149.3 ms PLL, including two skipped singletons. More singletons skip both stages in the population, explaining the smaller weighted increments. Native HF skips rescoring, trading faster decoding for 1.46 pp higher FLEURS WER and over 2.1 pp on CV27 TEST.

H3Direct uses H3 off-trigger and HF on triggers, omitting PLL. On CV27 it saves 6.29% over Direct but trails Native HF by 55.2 ms (95% interval 52.3 to 58.2 ms); the FLEURS interval spans zero. H4 and HWR4 only replace G4's weights; selective PLL remains future work. In Study B, Direct skips scoring on the CV27 trigger responsible for B's peak CUDA memory. On FLEURS, Direct and H3Direct match S+R's peak CUDA allocation when both windows' states are retained during recovery.

TABLE IV
TEST B WER (%) AND GAIN B−(S+R) (PP), AVERAGED OVER SEEDS BEFORE ROUNDING. CALLS POOL SEED–RECORDINGS.

Language	FLEURS			CV27		
	B	Gain	Calls, %	B	Gain	Calls, %
Arabic	24.84	0.64	0.23	52.12	1.40	0.11
Persian	31.08	1.20	0.27	69.02	9.05	1.15
Polish	13.57	0.00	0.00	14.57	0.30	0.33
Turkish	11.67	0.00	0.00	20.52	0.51	0.03
Italian	6.11	0.00	0.00	13.66	0.23	0.02
Empty-HF acceptance diagnostic (18 instances, 11 recordings)						
Acceptance minus B reversion	$S = -118$		$D = +120$		$I = -1995$	

Only HF proposes empty TEST candidates. Their zero-feature reference outranks nonempty candidates with $a, p \leq 0$, $t > 0$ and $r \geq 0$ under signs $(+, +, -, -)$, even without repetition. The 18 acceptances in 11 CV27 recordings remove 1,995 insertions and add 120 deletions (Table IV), consistent with hallucination suppression [33] but without verified silence labels. Reverting them to B retains 91.52% of gain, or 2.10 pp at 31.88% WER. H4 refitted with $e(y) = \mathbf{1}\{y = \emptyset\}$ on the same DEV folds and grid retains all 18 acceptances at 31.52% WER. One or two DEV empties per checkpoint limit calibration.

Persian has the highest CV27 baseline WER (71.61% raw, 69.02% B), contributing 78.77% of TEST gain and 84.07% on the locked panel. Compression is script-sensitive, but encoding and acoustic effects on these rates are not isolated. Excluding Persian, four-language macro gains remain 0.61 pp on CV27 TEST and 0.45 pp on the locked panel. Polish, Turkish and Italian have no FLEURS triggers or gains. The 22 two-window FLEURS recordings yield 52.69% of net gain. On cached TEST banks, local S+R reduces window requests from 15 to 10 without changing errors. Local Direct changes WER by under 0.01 pp, though live request order may alter RNG draws.

V. CONCLUSION

Selective acquisition matches unconditional recovery across seeds on the locked confirmation panel. It separates sparse recovery from routine rescoring to reduce recovery overhead. Retrospective controls favor expected risk over unweighted ranking on matched HF banks in every seed.

The choice of acceptance should reflect fallback screening. Reselection filters unscreened draws; Direct bypasses reselection of screened HF outputs. HWR4 gives the lowest exploratory CV27 WER, with gains that vary by checkpoint. Window-local routing and confidence-gated PLL could reduce acquisition and rescoring costs.

ACKNOWLEDGMENT

Supported by the National Natural Science Foundation of China under Grant 62466058 and Autonomous Region Key R&D Program under Grant 2025B03042-1.

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